

w12L56: Sentiment Analysis - Introduction

Example: Positive or negative movie review?



- unbelievably disappointing



- Full of zany characters and richly applied satire, and some great plot twists



- this is the greatest screwball comedy ever filmed



- It was pathetic. The worst part about it was the boxing scenes.

Where is Sentiment Analysis Used?

Movie Is this review positive or negative?

Products What do people think about the new iPhone?

Public Sentiment How is consumer confidence? Is despair increasing?

Politics What do people think about this candidate or issue?

Prediction Predict election outcomes or marked trends from sentiment

Where is Sentiment Analysis Used?

- Frustration of callers to a help line
- Stress in drivers or pilots
- Depression and other medical conditions from social media
- Confusion in students talking to e-tutors

Affective States Typology

Emotion: angry, sad, joyful, fearful, ashamed, proud, elated

Mood: cheerful, gloomy, irritable, listless, depressed, buoyant

Interpersonal stances: friendly, flirtatious, distant, cold, warm, supportive, contemptuous

Attitudes: liking, loving, hating, valuing, desiring

Personality Traits: nervous, anxious, reckless, morose, hostile, jealous

Sentiment Analysis

Sentiment Analysis is the detection of attitudes

enduring, affectively colored beliefs, dispositions towards objects or persons

The complete task

- Holder (source) of attitude
- Target (aspect) of attitude
- Type of attitude
 - From a set of types: *like, love, hate, value, desire*
 - Or simple weighted polarity: *positive, negative, neutral, together with strength*
- Text containing the attitude

Sentiment Analysis

Simplest Task

Is the attitude of this text positive or negative?

More complex

Rank the attitudes of this text from 1 to 5

Advanced

Detect the target, source, or complex attitude types

Sentiment Analysis in Movie Reviews

Polarity detection

Is an IMDB movie review positive or negative?



when _star wars_ came out some twenty years ago , the image of traveling throughout the stars has become a commonplace image . [...]
when han solo goes light speed , the stars change to bright lines , going towards the viewer in lines that converge at an invisible point .
cool .
october sky offers a much simpler image—that of a single white dot , traveling horizontally across the night sky . [. . .]



“ snake eyes ” is the most aggravating kind of movie : the kind that shows so much potential then becomes unbelievably disappointing .
it’s not just because this is a brian depalma film , and since he’s a great director and one who’s films are always greeted with at least some fanfare .
and it’s not even because this was a film starring nicolas cage and since he gives a brauvara performance , this film is hardly worth his talents .

Baseline Algorithm

- Tokenization
- Feature Extraction
- Classification using different classifiers
 - Naive Bayes
 - MaxEnt
 - SVM

Tokenization Issues

- Capitalization - preserve for word in all caps
- Word lengthening
- Handling emoticons

```
[<>]?          # optional hat/brow
[:;=8]         # eyes
[\-o\*\ ' ]?   # optional nose
[\)\)\]\(\[dDpP/\:}\{\@\|\|\]  # mouth
|              ##### reverse orientation
[\)\)\]\(\[dDpP/\:}\{\@\|\|\]  # mouth
[\-o\*\ ' ]?   # optional nose
[:;=8]         # eyes
[<>]?          # optional hat/brow
```

- Handling negation
 - I **didn't** like this movie
 - I really like this movie
- Add NOT_ to every word between negation and following punctuation
 - *didn't like this movie, but I ...*
 - didn't NOT_like NOT_this NOT_movie but I..

Naive Bayes: Reminder

$$c_{NB} = \arg \max_{c_j \in C} P(c_j) \prod_{x_i} P(x_i | c_j)$$

$$\hat{P}(c_j) = \frac{\text{doc} - \text{count}(C = c_j)}{N_{\text{doc}}}$$

$$\hat{P}(w_i | c_j) = \frac{\text{count}(w_i, c_j) + 1}{(\sum_{w \in V} (\text{count}(w, c_j)) + |V|)}$$

Boolean Multinomial Naive Bayes

First remove all duplicate words from a test document d
Then compute NB using the same equation

$$c_{NB} = \arg \max_{c_j \in C} P(c_j) \prod_{x_i} P(x_i | c_j)$$

A piece of cake?

Is a given review on a known topic positive or negative?

“It may be a bit early to make such judgments, but Battlefield Earth may well turn out to be the worst movie of this century.” (Elvis Mitchell, May 12, 2000)

“It may be a bit early to make such judgments, but Battlefield Earth may well turn out to be the worst movie of this century.” (Elvis Mitchell, May 12, 2000)

don't we just need to look for “worst”, “best”, “love”, “hate”, etc.?

In a small scale experiment (Pang et al., 2002)

	Proposed word lists	Accuracy
Human 1	Positive: dazzling, brilliant, phenomenal, excellent, fantastic Negative: suck, terrible, awful, unwatchable, hideous	58%
Human 2	Positive: gripping, mesmerizing, riveting, spectacular, cool, awesome, thrilling, badass, excellent, moving, exciting Negative: bad, cliched, sucks, boring, stupid, slow	64%
Statistics-based	Positive: love, wonderful, best, great, superb, beautiful, still Negative: bad, worst, stupid, waste, boring, ?, !	69%

Why can't we just look for words like “great” and “terrible”?

- This laptop is a **great deal**.
 - This is positive
- A **great deal** of media attention surrounded the release of the new laptop.
 - It is neutral - says nothing about laptop
- This laptop is a *great deal* ... and I've got a nice bridge you might be interested in.
 - Sarcastic - really bad laptop - like a joke
- This film should be brilliant. It sounds like a great plot, the actors are first grade, and the supporting cast is good as well, and Stallone is attempting to deliver a good performance.
However, it can't hold up.
 - Last sentence turns up the polarity to negative

Have a large list of Known Words - Affective Lexicons!

W12L2:Sentiment Analysis - Affective Lexicons

Sentiment Lexicons

- The **general inquirer** (popular lexicon)
- **MPQA** Subjectivity Cues Lexicon (Multi Perspective Q&A)
- **SentiWordnet** (used widely)
- **LIWC** (Linguistic Inquiry and Word Count) - recently used for sentiment analysis

1. The General Inquirer - *Categories - comes in spreadsheet*

- Positive (1915 words) and Negative (2291) words
- Dimension: Strong vs weak, active vs passive, overstated vs understated
 - pleasure, pain, virtue, vice, motivation, cognitive orientation etc.

Entry	Source	Positiv	Negativ	Pstv	Affil	Ngvtv	Hostile	Strong	Power	Weak	Submit	Active	Passive
A	H4Lvd												
ABANDON	H4Lvd		Negativ			Ngvtv				Weak			
ABANDONMENT	H4		Negativ							Weak			
ABATE	H4Lvd		Negativ										Passive
ABATEMENT	Lvd												
ABDICATE	H4		Negativ							Weak	Submit		Passive
ABHOR	H4		Negativ				Hostile						Passive
ABIDE	H4	Positiv			Affil							Active	
ABILITY	H4Lvd	Positiv						Strong					
ABJECT	H4		Negativ							Weak	Submit		Passive
ABLE	H4Lvd	Positiv		Pstv				Strong					
ABNORMAL	H4Lvd		Negativ			Ngvtv							
ABOARD	H4Lvd												
ABOLISH	H4Lvd		Negativ			Ngvtv	Hostile	Strong	Power			Active	
ABOLITION	Lvd												
ABOMINABLE	H4		Negativ					Strong					
ABORTIVE	Lvd												
ABOUND	H4	Positiv											Passive
ABOUT#1	H4Lvd												
ABOUT#2	H4Lvd												
ABOUT#3	H4Lvd												
ABOUT#4	H4Lvd												
ABOUT#5	H4Lvd												
ABOUT#6	H4Lvd												
ABOUT#7	H4Lvd												
ABOVE#1	H4Lvd												
ABOVE#2	H4Lvd												
ABOVE#3	H4Lvd												
ABOVE#4	H4Lvd												
ABRASIVE	H4		Negativ				Hostile	Strong					
ABROAD	H4Lvd												
ABRUPT	H4Lvd		Negativ			Ngvtv							
ABSCOND	H4		Negativ				Hostile			Weak	Submit	Active	
ABSENCE	H4Lvd		Negativ							Weak			
ABSENT#1	H4Lvd		Negativ							Weak			Passive
ABSENT#2	H4Lvd												Passive
ABSENT-MINDED	H4		Negativ							Weak			Passive

2. SentiWordNet - Home page: <http://sentiwordnet.isti.cnr.it/>

All Wordnet **synsets** automatically annotated for degrees of - 3 sentiment scores

1. positivity,
2. negativity, and
3. neutrality/objectiveness

Example

estimable (J,3) : “may be computed or estimated”

Pos 0 Neg 0 Obj 1

estimable (J,1) : “deserving of respect or high regard”

Pos .75 Neg 0 Obj .25

Other Lexicons

3. MPQA Subjectivity Cues Lexicon - Home page: http://www.cs.pitt.edu/mpqa/subj_lexicon.html

- 6885 words from 8221 lemmas: 2718 positive, 4912 negative
- Each word annotated for **intensity** (strong, weak)

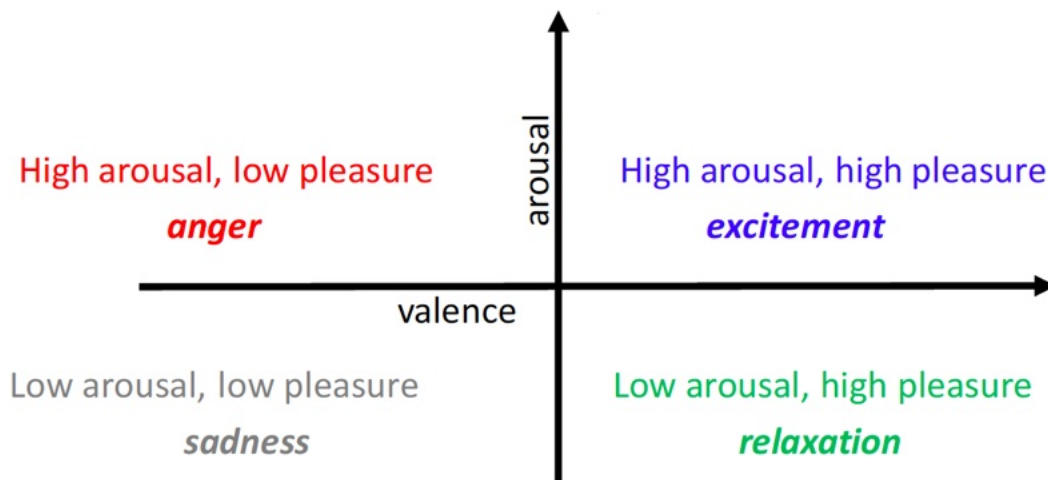
4. Bing Liu Opinion Lexicon - Bing Liu's Page on Opinion Mining

- <http://www.cs.uic.edu/~liub/FBS/opinion-lexicon-English.rar>
- 6786 word: 2006 positive, 4780 negative

5. LIWC (Linguistic Inquiry and Word Count) - Home page: <http://www.liwc.net/>

- 2300 words, > 70 classes
 - Simple +ve, -ve to social things
 - Classes make richer representations - can be used for sentiment analysis
 - Emotion, interpersonal things etc.
- Comes with a small fee
- **Affective Processes**
 - Negative emotion (bad, weird, hate, problem, tough)
 - Positive emotion (love, nice, sweet)
- **Cognitive Processes**
 - Tentative (maybe, perhaps, guess), Inhibition (block, constraint)

Valence (+ve or -ve) / Arousal (strength/ weak) Dimensions



Valence : -ve to +ve

Arousal: Low to High

Lexicon of valence, arousal, and dominance

- Warriner, Amy Beth, Victor Kuperman, and Marc Brysbaert. “*Norms of valence, arousal, and dominance for 13,915 English lemmas*.” Behavior research methods 45.4 (2013): 1191-1207.
- **Supplementary data:** This word is licenced under a Creative Commons.
 - Available for non-commercial use

Ratings for 14,000 words for emotional dimensions:

- **valence** (the **pleasantness** of the stimulus)
- **arousal** (the **intensity** of emotion provoked by the stimulus)
- **dominance** (the degree of **control** exerted by the stimulus)

Lexicon of valence, arousal, and dominance

valence (the pleasantness of the stimulus)

- **9:** happy, pleased, satisfied, contented, hopeful
- **1:** unhappy, annoyed, unsatisfied, melancholic, despaired, or bored

arousal (the intensity of emotion provoked by the stimulus)

- **9:** stimulated, excited, frenzied, jittery, wide-awake, or aroused
- **1:** relaxed, calm, sluggish, dull, sleepy, or unaroused

dominance (the degree of control exerted by the stimulus)

- **9:** in control, influential, important, dominant, autonomous, or controlling
- **1:** controlled (by someone else), influenced, cared-for, awed, submissive, or guided

Lexicon of valence, arousal, and dominance: Examples

Valence		Arousal		Dominance	
vacation	8.53	rampage	7.56	self	7.74
happy	8.47	tornado	7.45	incredible	7.74
whistle	5.7	zucchini	4.18	skillet	5.33
conscious	5.53	dressy	4.15	concur	5.29
torture	1.4	dull	1.67	earthquake	2.14

Some of these were created manually

- And verified

Can we learn these lexicons from data

- Bootstrap

w12L58: L12W3: Learning Affective Lexicons

Learning Sentiment Lexicons

Basic Intuition

- When do the words of same or opposite polarity occur together?
- Adjectives conjoined by “**and**” have same polarity
 - *Fair and legitimate, corrupt and brutal*
- Adjectives conjoined by “**but**” do not
 - *fair but brutal*

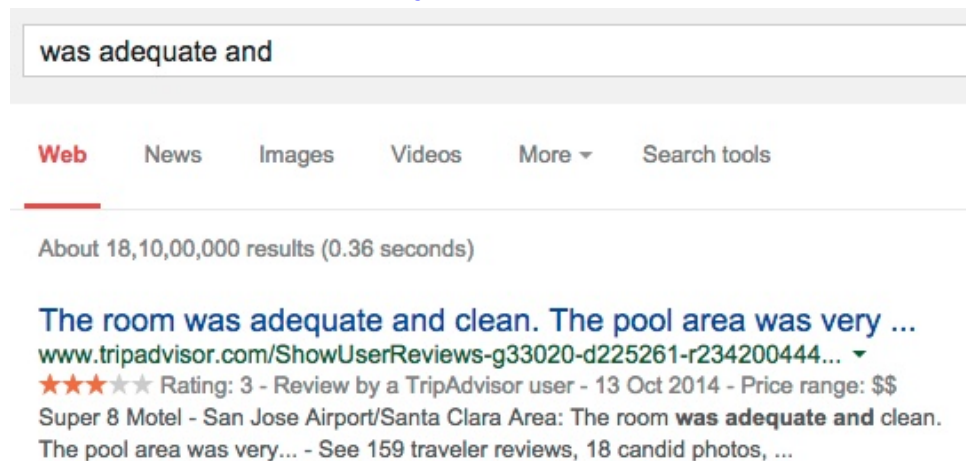
From web:

1st approach: take fair. Find fair and x, fair but x

Step 1: Label seed set of adjectives

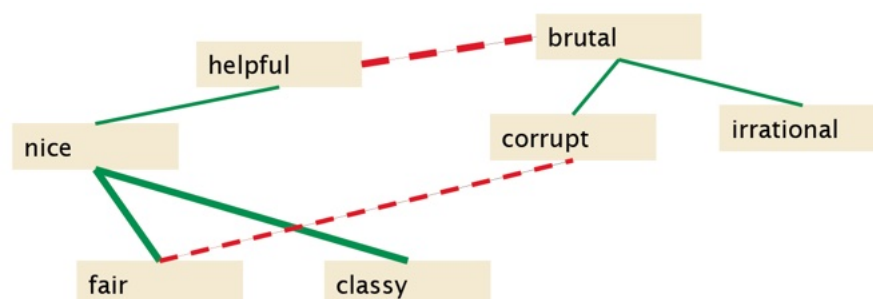
- **Positive cases:** adequate, central, clever, famous, intelligent, remarkable, reputed, sensitive, slender, thriving ...
- **Negative cases:** contagious, drunken, ignorant, lanky, listless, primitive, strident, troublesome, unresolved, unsuspecting ...

Step 2: Expand seed set to conjoined adjectives - search on web



Step 3: Construct a graph

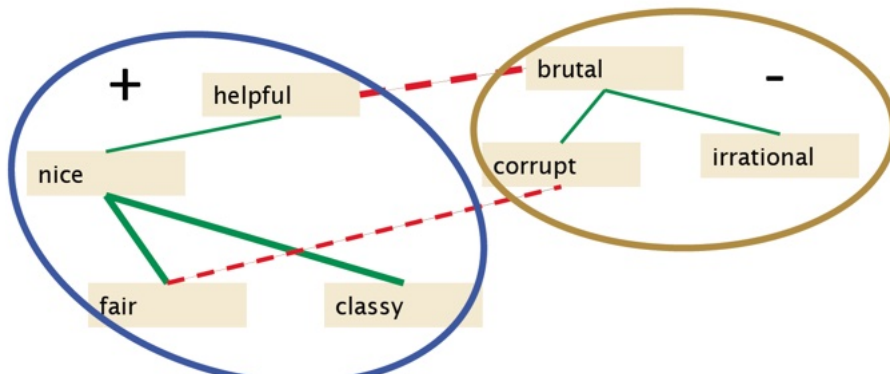
Polarity similarity is assigned to each word pair: nice-fair, fair-classy occur together



You label edges: positive and negative polarity - green, red edges

Clustering for partitioning the graph into two

- Do it using seed set and large corpus
- You can find other rule - not just and or but



Output Polarity Lexicon

Positive - you will get noisy examples

bold decisive disturbing generous good honest important large mature patient peaceful positive proud sound stimulating straightforward strange talented vigorous witty ...

Negative -

ambiguous cautious cynical evasive harmful hypocritical inefficient insecure irrational irresponsible minor outspoken pleasant reckless risky selfish tedious unsupported vulnerable wasteful ...

Positive - erroneous cases - bold

bold decisive **disturbing** generous good honest important large mature patient peaceful positive proud sound stimulating straightforward strange talented vigorous witty ...

Negative - erroneous cases - bold

ambiguous **cautious** cynical evasive harmful hypocritical inefficient insecure irrational irresponsible minor **outspoken pleasant** reckless risky selfish tedious unsupported vulnerable wasteful ...

1. Turney Algorithm

Step 1:

- Start with dataset - let us take reviews dataset
 - Adjectives and Nouns
 - Extract Noun phrases
 - Find sentiment of Noun
- Extract a **phrasal** lexicon from reviews
- Learn **polarity** of each phrase
- Rate a review by the average polarity of its phrases

Extract two-word phrases with adjectives

- How sentiment phrases occur
- Built patterns by looking at the data

First Word	Second Word	Third Word (not extracted)
JJ	NN or NNS	anything
RB, RBR, RBS	JJ	Not NN nor NNS
JJ	JJ	Not NN or NNS
NN or NNS	JJ	Nor NN nor NNS
RB, RBR, or RBS	VB, VBD, VBN, VBG	anything

Step 2: How to give sentiment?

Measuring the polarity of the phrases

- Positive phrases co-occur more with “excellent”
- Negative phrases co-occur more with “poor”
- How to measure the co-occurrence?

Simple count will not work?

How common two words cooccur?

Pointwise Mutual Information (PMI):

$$PMI(X, Y) = \log_2 \frac{P(x, y)}{P(x)P(y)}$$

How to estimate PMI?

Query search engine (Altavista)

- $P(\text{word})$ estimated by $\text{hits}(\text{word})/N$
- $P(\text{word}_1, \text{word}_2)$ estimated by $\text{hits}(\text{word}_1 \text{ NEAR } \text{word}_2)/N$

Polarity(phrase) = PMI(phrase, excellent) - PMI(phrase, poor)

$$= \log_2 \left(\frac{\text{hits}(\text{phrase NEAR "excellent"})\text{hits}(\text{"poor"})}{\text{hits}(\text{phrase NEAR "poor"})\text{hits}(\text{"excellent"})} \right)$$

$$PMI(\text{phrase}, \text{excellent}) - PMI(\text{ph}, \text{poor})$$

$$= \log \frac{P(\text{phrase}, \text{excellent})}{P(\text{phrase}) \cdot P(\text{excellent})} \cdot \frac{P(\text{ph})P(\text{poor})}{P(\text{ph}, \text{poor})}$$

$$PMI(\text{phrase}, \text{excellent}) - PMI(\text{ph}, \text{poor})$$

$$= \log \left[\frac{P(\text{phrase}, \text{excellent})}{P(\text{phrase}) \cdot P(\text{excellent})} \cdot \frac{P(\text{ph})P(\text{poor})}{P(\text{ph}, \text{poor})} \right]$$

hits (phrase near excell)

Example: A thumbs-up Review

Phrase	POS tags	Polarity
online service	JJ NN	2.8
online experience	JJ NN	2.3
direct deposit	JJ NN	1.3
local branch	JJ NN	0.42
...		
low fees	JJ NNS	0.33
true service	JJ NN	-0.73
other bank	JJ NN	-0.85
inconveniently located	JJ NN	-1.5
Average		0.32

Example: A thumbs-down Review

Phrase	POS tags	Polarity
direct deposits	JJ NNS	5.8
online web	JJ NN	1.9
very handy	RB JJ	1.4
...		
virtual monopoly	JJ NN	-2.0
lesser evil	RBR JJ	-2.3
other problems	JJ NNS	-2.8
low funds	JJ NNS	-6.8
unethical practices	JJ NNS	-8.5
Average		-1.2

Thumbs up: What is the average = 0.32 (there were some -ve examples e.g. -1.5)

Thumbs down: Average = -1.2

2. Using WordNet to learn polarity

- Find positive and negative words
 - Find out all the synonyms
- WordNet: online thesaurus
- Create positive (“good”) and negative seed-words (“terrible”)
- Find Synonyms and Antonyms
 - Positive Set: Add **synonyms** of **positive** words (“well”) and **antonyms** of **negative** words
 - Negative Set: Add **synonyms** of **negative** words (“awful”) and **antonyms** of **positive** words (“evil”)
- **Repeat**, following chains of synonyms
- Filter

w12L59: W12L4 Computing with Affective Lexicons

What things we need to keep in mind

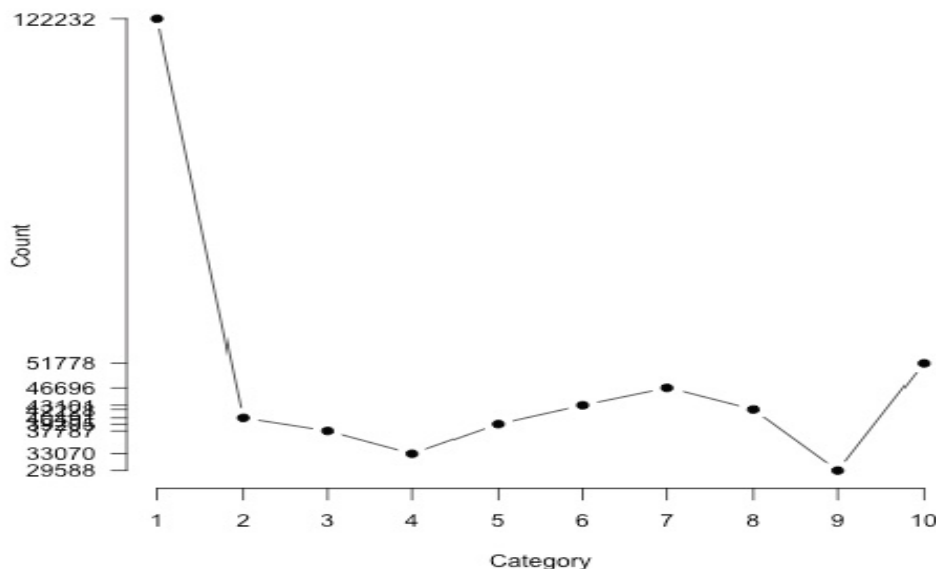
- What kind of measure use to find trend or are prominent for positive or negative

Learn word sentiment supervised by online review scores

- Review datasets: **Movies**-IMDB, Goodreads, **Amazon**, Trip Advisor
- Each review has a score (1-5, 1-10 etc)
- Just **count** how many times each word occurs with each **score** (and normalize).

Analyzing polarity of each word in IMDB

- How likely is each word to appear in each sentiment class?
- Let's take count("bad") in 1-star, 2-star, 3-star etc.



Can not help in finding normalised structure?

- Does it occur with more probability?
- I have more review of rating with 1 than rating 5

- We should use likelihood instead of counts: $P(w|c) = \frac{f(w,c)}{\sum_{w \in c} f(w,c)}$

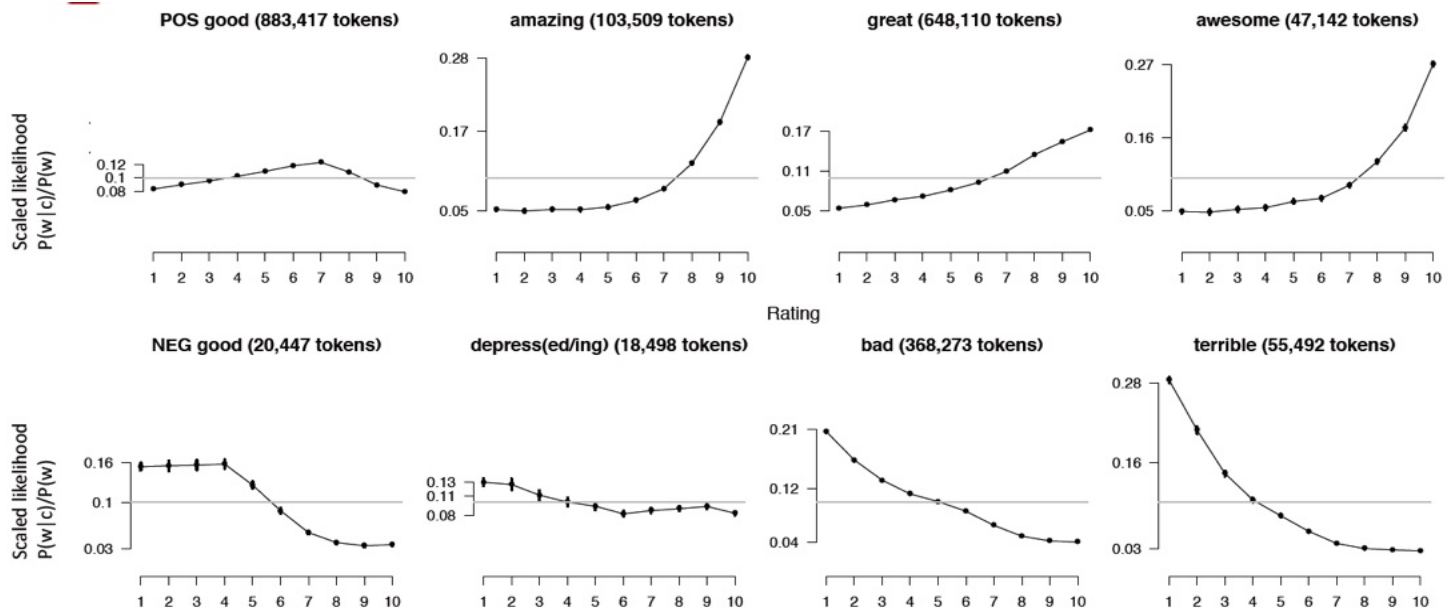
Likelihood: Will help normalise across the rating

- Make them comparable between words Scaled likelihood: $\frac{P(w|c)}{P(w)}$

Scaled Likelihood: To compare across words - word is very common - prob word is very high, so occurring it with rating with high

- So again normalise - it will be comparable with other words
- How much does it depart from actual probability

Analyzing polarity in IMDB



X-axis - rating, y-axis scaled Likelihood

Figure col1-top: it is increasing then decreasing.

figure col1-bottom: -ve goods is high then decreasing - even higher reviews - you may use amazing, fantastic

Col2-top: Amazing: for positive goes up

Col-4: top

Col3-top: great - increasing - not as steep as amazing, terrible has similar trend

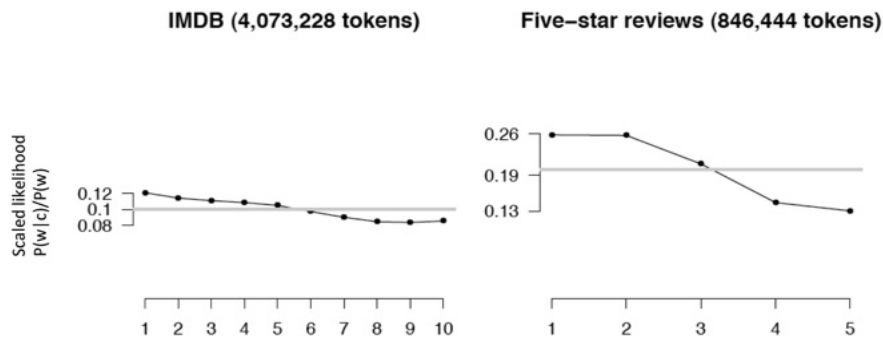
You can compare different words - which words behave in similar manner

- Can use for which words are +ve and -ve

Logical Negation

- Is logical negation (no, not) associated with negative sentiment?
- **Potts** experiment:
 - **Count** negation (not, n't, no, never) in online reviews
 - **Regress** against the review rating

More negation in negative sentiment



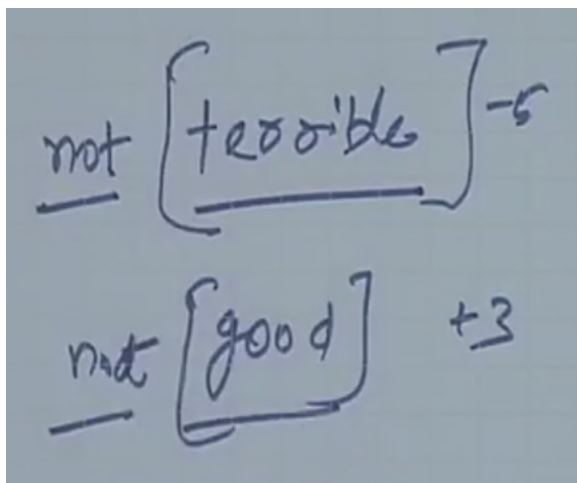
Review of rating 1: Likelihood - negation is occurring more in reviews with rating 1 than 2
Review rating 5: likelihood 0.26 than review rating 5 0.13

Baseline:

What is the sentiment of each word and take sentence?

Better:

- Include linguistic features



If Not is there then we reverse the polarity

- Very good ->
- Some what good ->

Use this to make this linguistic intuition

Using Linguistic Intuitions

Using a sentiment lexicon also works.

Some linguistic intuitions on top of that tends to give better results.

Handling negation in simple addition of scores

Example words	Reversing the polarity
• Excellent +5 • good +3 • terrible -5 • bad -3	• Not Excellent -5 • Not good -3 • Not terrible +5 • Not bad +3

Not excellent -> may still be good -> changing to -5 may not be ideal

You want to give not good lower scale

Not terrible should have higher score than Not good

Handwritten calculations showing the process of reversing polarity for negated words:

$$\text{not } [\text{terrible}] \quad -5 + 4 = -1$$

not bad

$$\text{not } [\text{good}] \quad +3 \quad -3 + 4 = 1$$

Handwritten diagram and calculations illustrating the process of reversing polarity for negated words:

Diagram showing a box labeled "Not Excellent -5" and "Not good -3".

Calculations:

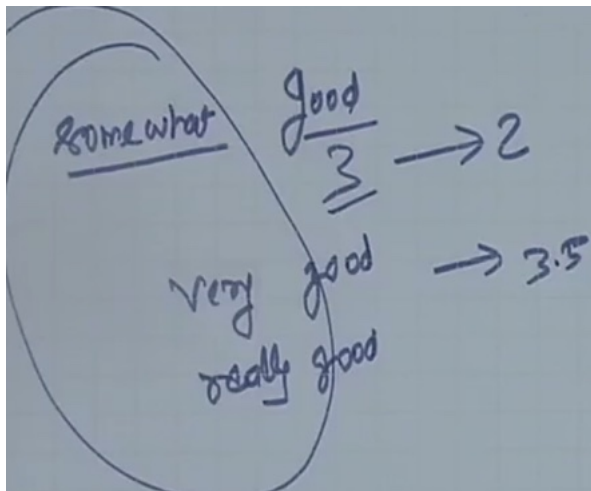
$$\text{EX. } 5 \quad 5 - 4 = 1$$
$$\text{not ex } 3 - 4 = -1$$

Additional notes: "not", "not good", "not", "not good".

Use Polarity Shifter:

Example words • Excellent +5 • good +3 • terrible -5 • bad -3	Instead, a polarity shift works better • Not Excellent (5-4) +1 • Not good (3-4) -1 • Not terrible (-5+4) -1 • Not bad (-3+4) 1
--	--

Not good is more -ve than Not Excellent



acting as function words

Handling Intensifiers (use amplifiers)

Intensifiers can be classified into two major categories,

Amplifiers (e.g., very) increase the semantic intensity

Downtoners (e.g., slightly) decrease it

Rough values for some intensifiers

Intensifier	Modifier (%)
slightly	-50
somewhat	-30
pretty	-10
really	+15
very	+25
extraordinarily	+50
(the) most	+100

Sentence is: Somewhat sleazy

sleazy: -3, somewhat sleazy: $-3 \times (100\% - 30\%) = -2.1$

Irrealis moods: where the words may not be reliable

- I thought this movie would be as good as the Grinch, but unfortunately, it wasn't.
- Can rely on sentiment analysis
- This should have been a great movie. (it doesn't mean positive)

What are the indicators?

- conditional markers (**if**)
- **negative** polarity items like '**any**' and '**anything**'
- certain (mostly **intentional**) verbs (expect, doubt),
- **questions**
- words enclosed in **quotes** (which may be factual, but not necessarily reflective of the author's opinion)
 - Can't assign sentiment based on sentence in quote
 - Don't use it

Refer: Sentiment Tutorial by Christopher Potts

Text scoring

Shows how a variety of **sentiment lexicons** score novel texts. Such values could be used in many ways (as raw values, to derive percentages or ratios, as **classifier features**, ...).

Enter your own text (max 140 characters), or

It sounds like a great plot but can't hold up.

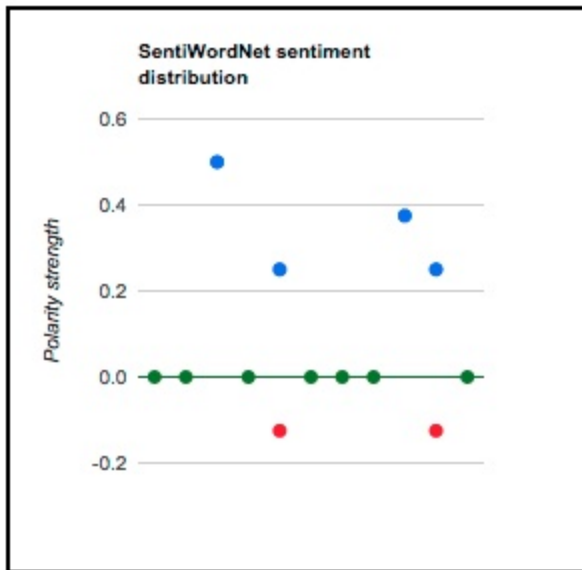
☐ analyze a random tweet instead.

Submit

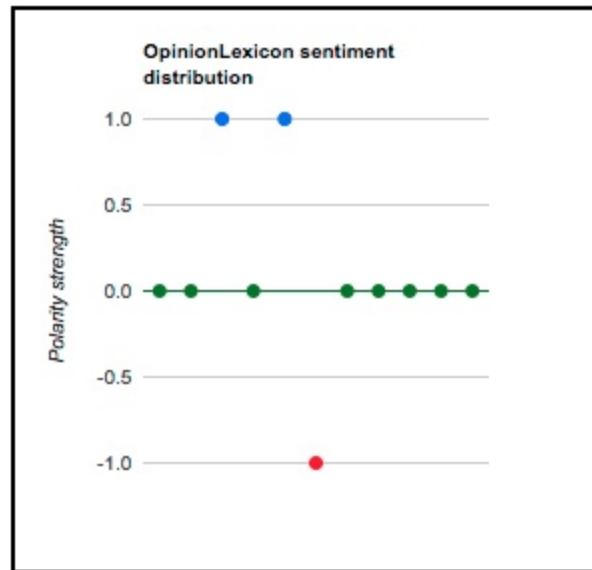
Scores

it sounds like a great plot but can't hold_NEG up_NEG .

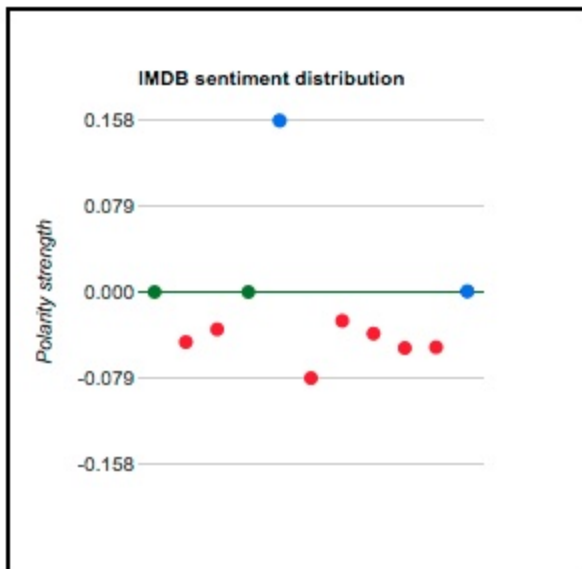
SentiWordNet



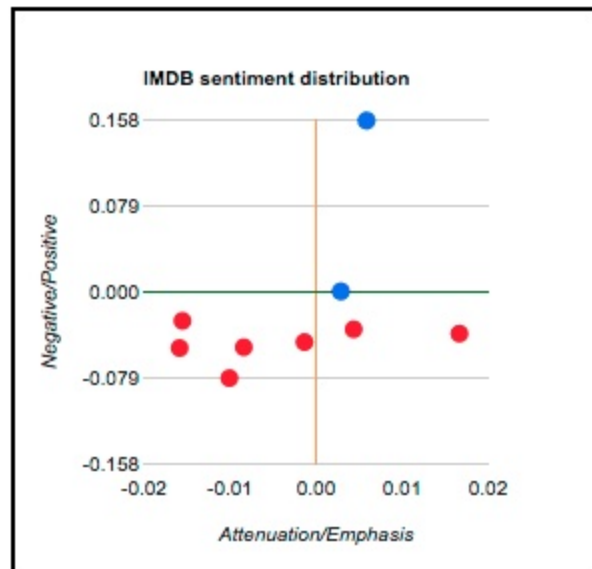
Opinion Lexicon



IMDB



IMDB 2d



w12L60: W12L5: Aspect-based Sentiment Analysis

Finding **aspects** or **attributes**

- In Camera product review:
- You can use to compare product - resolution, picture quality

Sentence - In Hotel review: : The food was great but the service was awful.

Aspects Involved (can we capture)

- Food, service

What kind of noun phrase and occur lot directly after sentiment word?

Approach 1: Finding aspect/ attribute/ target of sentiment

Frequent phrases + rules

- Find all highly frequent phrases across reviews (“fish tacos”)
- Filter by rules like “**occurs right after sentiment word**”
- “... great fish tacos” means “fish tacos” a likely aspect

Casino	casino, buffet, pool, resort, beds
Children’s Barber	haircut, job, experience, kids
Greek Restaurant	food, wine, service, appetizer, lamb
Department Store	selection, department, sales, shop, clothing

Sometimes have aspects - specific to domains

Approach 2: Extraction of aspect-opinion pairs (using *dependency parse*)

RuleID	Observations	Examples
R1	$JJ \leftarrow amod \leftarrow NP$	The camera has a good screen. (good $\leftarrow$ amod $\leftarrow$ screen)
R2	$NP \rightarrow nsubj \rightarrow JJ$	The flash is brilliant. (flash $\leftarrow$ nsubj $\leftarrow$ brilliant)
R3	$VB \rightarrow dobj \rightarrow NP$	I love the image quality. (love $\rightarrow$ dobj $\rightarrow$ image quality)
R4	$NP \rightarrow nsubj \rightarrow JJ,$ $JJ \in \text{implicit aspect lexicon}$	The camera is expensive. (camera $\leftarrow$ nsubj $\leftarrow$ expensive)

R1: JJ is opinion, NP is aspect

Expensive -> is implicit aspect -> talks about Price

Once you have aspects, then use classifier

- If sentence talks about one of the aspects

Finding aspect/attribute/target of sentiment

If the aspects are well understood, use supervised classification (NB, LR, etc.)

- Label different sentences
- For new sentence find the aspect

Rooms (3/5 stars, 41 comments)

(+) The room was clean and everything worked fine – even the water pressure ...

(+) We went because of the free room and was pleasantly pleased ...

(-) ...the worst hotel I had ever stayed at ...

Service (3/5 stars, 31 comments)

(+) Upon checking out another couple was checking early due to a problem ...

(+) Every single hotel staff member treated us great and answered every ...

(-) The food is cold and the service gives new meaning to SLOW.

Dining (3/5 stars, 18 comments)

(+) our favorite place to stay in biloxi.the food is great also the service ...

(+) Offer of free buffet for joining the Play

Do opinion phrases always have the same sentiment?

'Large' – positive or negative

Large screen vs. Large battery (becoming heavy)

'Long' – positive or negative

Long battery life vs. Long loading time

Have to talk about opinion wrt to aspect

Explicit vs. Implicit Aspect Expressions

- *The picture quality of this camera is great* – 'picture quality' is an **explicit** aspect.
- *This camera is expensive* – 'expensive' is an **implicit** aspect expression describing 'price'.
- **Implicit** aspect expressions can be very **complex** as well, e.g., *This camera will not fit in a pocket* – "fit in a pocket" indicates the aspect **'size'**.

Finding Implicit Aspect: interesting application

1. Interesting Applications

What are the aspects - what are the people's comment on

- price or picture (image quality) - Taking Opining and Trying to Summarize

Aspect-based Opinion Summarization

"I bought an *iPhone* a few days ago. It is such a nice *phone*. The *touch screen* is really cool. The *voice quality* is clear too. It is much better than my old *Blackberry*, which was a terrible *phone* and so difficult to type with its *tiny keys*. However, my *mother* was mad with me as I did not tell her before I bought the *phone*. She also thought the phone was too *expensive*, ..."

Feature Based Summary of iPhone:

Feature1: Touch screen

Positive: 212

- The *touch screen* was really cool.
- The *touch screen* was so easy to use and can do amazing things.

...

Negative: 6

- The *screen* is easily scratched.
- I have a lot of difficulty in removing finger marks from the *touch screen*.

...

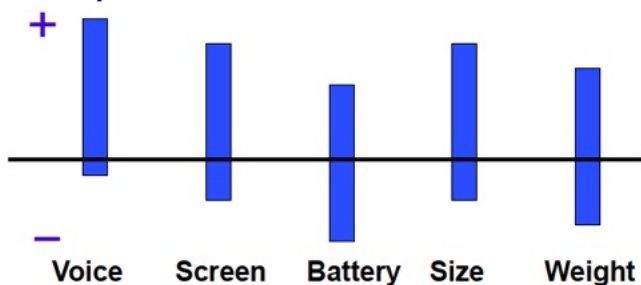
Feature2: voice quality

2. Interesting Applications

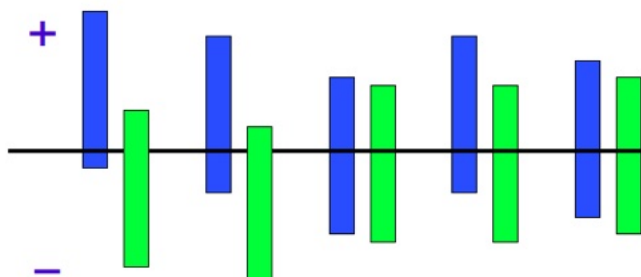
- Compare product based on spec
- Not comparing - voice quality
 - We can talk about summary, better because customers are saying
 - Why should you go for product 1 vs product 2

Aspect-based Product Comparison

- Summary of reviews of
- Cell Phone 1



- Comparison of reviews of
- Cell Phone 1
- Cell Phone 2



Conclusion

Many more NLP Applications

- Machine Translation
 - Align words in two sentences
- Question Answering
 - Have lot of text, you want actual answers
 - Challenging
- Cross-lingual Applications
 - Information in Hindi (is typed in roman)
- Text Processing for social media, informal text
 - Process language of social media - informal text
- Code switching and code mixing
 - Start with english -> hindi (completely)
 - Mixing -> you mix words
 - Sentiment analysis
 - POS tagging

Deep Learning Techniques are being applied for most of the tasks.

Exam Format:

20 Questions: 1 mark each -> recall type questions (from slides)

20 questions: 2 marks each -> based on assignments (Assignments)

8 questions: 5 marks each -> Not covered in assignment, need to give some more time

Most questions will have single answer

For 2-3 question - you may have to give numeric values

Great Job!